

Toolken+: Improving LLM Tool Usage with Reranking and a Reject Option

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Goals of the research

Problem

Tool learning with Large Language Models.

Challenges

First, ToolkenGPT cannot use tool documentation known to be helpful for LLMs. Second, ToolkenGPT often makes mistakes in judging when to use tools, calling them too often.

Solutions

Resolve the first challenge by re-ranking the retrieved tools and the second challenge by introducing an additional "Reject" tool.

Background

Augmented Large Language Models

Fine-tuning. Using a standard language modelling objective, fine-tune the LLM on structured data of the form:

Out of 1400 participants, 400 (or `[Calculator(400 / 1400) → 0.29]` 29%) passed the test.

In-context learning. LLMs learn to utilize tools through in-context demonstrations that are embedded within the prompts.

Paradigms	Frozen LLM	Massive Tools	Plug-&-Play	Extensive Data
Fine-tuning	✗	✗	✗	✓
In-context learning	✓	✗	✓	✗
ToolkenGPT	✓	✓	✓	✓

Background

ToolkenGPT

Given the embedding matrix $\mathbf{W}_{\mathcal{V}} \in \mathbb{R}^{|\mathcal{V}| \times d}$, where $\mathcal{V}$ is the original vocabulary tokens and the matrix $\mathbf{W}_{\mathcal{T}} \in \mathbb{R}^{|\mathcal{T}| \times d}$ for a set of tools $\mathcal{T}$. So, the next token probabilities are calculated as $p_{\text{aug}}(\mathbf{x}_i | \mathbf{x}_{<i}) = \text{softmax}([\mathbf{W}_{\mathcal{V}}, \mathbf{W}_{\mathcal{T}}] \mathbf{h}_{i-1})$.

Inference.

- 1 *Reasoning Mode*: where the model generates a rationale using $p_{\text{aug}}(\mathbf{x}_i | \mathbf{x}_{<i})$
- 2 *Tool Mode*: where it infills tool arguments given a prompt with usage examples

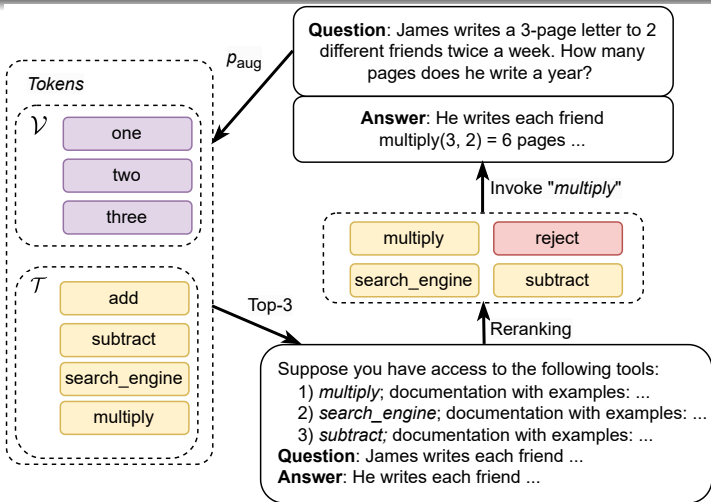
Training.

$$\min_{\mathbf{W}_{\mathcal{T}}} \sum_{X \in \mathcal{D}} \sum_{i=1}^{|X|} -\log p_{\text{aug}}(\mathbf{x}_i | \mathbf{x}_{<i}).$$

Challenges.

- ToolkenGPT cannot use tool documentation known to be helpful for LLMs.
- ToolkenGPT often makes mistakes in judging when to use tools.

Method: Toolken+



- extend the tool set as $\mathcal{T}' = \mathcal{T} \cup \{\text{REJ}\}$
- Toolken+ takes the retrieved subset of top- k tools $\mathcal{T}_k \subseteq \mathcal{T} \cup \{\text{REJ}\}$ and is asked to generate a tool from $\mathcal{T}_k \cup \{\text{REJ}\}$ using $p_{\text{rank}}(\mathbf{x}_i | \mathbf{x}_{<i})$

$$p_{\text{rank}}(\mathbf{x}_i | \mathbf{x}_{<i}, \mathcal{T}_k) = \text{softmax}(\mathbf{W}_{\mathcal{T}'} \mathbf{h}_{i-1} + \mathbf{m}(\mathcal{T}_k)), \quad \mathbf{m}(\mathcal{T}_k)_t = (-\infty) \cdot \llbracket t \notin \mathcal{T}_k \cup \{\text{REJ}\} \rrbracket$$

Algorithm 1: Toolken+ inference

Data: $p_{\text{LLM}}(\mathbf{x}_i | \mathbf{x}_{<i})$, $p_{\text{aug}}(\mathbf{x}_i | \mathbf{x}_{\leq i})$, $p_{\text{rank}}(\mathbf{x}_i | \mathbf{x}_{\leq i}, \mathcal{T}_k)$, user query q

Result: Rationale $\mathbf{x}_{1:n}$ (with a user query)

$x \leftarrow q$, $i \leftarrow |q|$;

while $\mathbf{x}_i \neq \text{EOS}$ **do**

$\mathbf{x}_{i+1}^{(0)} \sim p_{\text{aug}}(\cdot | \mathbf{x}_{\leq i})$;

if $\mathbf{x}_{i+1}^{(0)} \in \mathcal{T}$ **then**

$\mathcal{T}_k \leftarrow \text{TopkTools}(p_{\text{aug}}(\cdot | \mathbf{x}_{\leq i}))$;

$\mathbf{x}_{i+1}^{(1)} \sim p_{\text{rank}}(\cdot | \mathbf{x}_{\leq i}, \mathcal{T}_k)$;

if $\mathbf{x}_{i+1}^{(1)} = \text{REJ}$ **then** $\mathbf{x}_{i+1} \sim p_{\text{LLM}}(\cdot | \mathbf{x}_{\leq i})$ **else** $\mathbf{x}_{i+1} \leftarrow \mathbf{x}_{i+1}^{(1)}$;

else

$\mathbf{x}_{i+1} \leftarrow \mathbf{x}_{i+1}^{(0)}$;

$i \leftarrow i + 1$;

Approximate inference

Deriving a computationally stable criterion

Proposition 1 (next token probabilities)

The next token probabilities in Algorithm 1 are given by

$$\begin{aligned} p(\mathbf{x}_{i+1} | \mathbf{x}_{\leq i}) = & \\ & \llbracket \mathbf{x}_{i+1} \in \mathcal{V} \rrbracket (p_{\text{aug}}(\mathbf{x}_{i+1} | \mathbf{x}_{\leq i}) + p_{\text{aug}}(\mathcal{T} | \mathbf{x}_{\leq i}) p_{\text{rank}}(\text{REJ} | \mathbf{x}_{\leq i}, \mathcal{T}_k) p_{\text{LLM}}(\mathbf{x}_{i+1} | \mathbf{x}_{\leq i})) + \\ & \llbracket \mathbf{x}_{i+1} \in \mathcal{T} \rrbracket p_{\text{aug}}(\mathcal{T} | \mathbf{x}_{\leq i}) p_{\text{rank}}(\mathbf{x}_{i+1} | \mathbf{x}_{\leq i}, \mathcal{T}_k) \end{aligned}$$

The next token prediction loss is computationally unstable, so a new upper bound is proposed.

Theorem 1 (naive upper bound)

A computationally stable upper bound, up to an additive constant independent of $\mathbf{W}_{\mathcal{T}'}$, is

$$\sum_{X \in D} \sum_{i=1}^{|X|} (-\llbracket \mathbf{x}_{i+1} \in \mathcal{V} \rrbracket \log p_{\text{rank}}(\text{REJ} | \mathbf{x}_{\leq i}, \mathcal{T}_k) - \llbracket \mathbf{x}_{i+1} \in \mathcal{T} \rrbracket \log p_{\text{rank}}(\mathbf{x}_{i+1} | \mathbf{x}_{\leq i}, \mathcal{T}_k)).$$

Simplified objective

The bound

$$\sum_{X \in D} \sum_{i=1}^{|X|} \left(- \mathbb{I}[\mathbf{x}_{i+1} \in \mathcal{V}] \log p_{\text{rank}}(\text{REJ}|\mathbf{x}_{\leq i}, \mathcal{T}_k) - \mathbb{I}[\mathbf{x}_{i+1} \in \mathcal{T}] \log p_{\text{rank}}(\mathbf{x}_{i+1}|\mathbf{x}_{\leq i}, \mathcal{T}_k) \right)$$

is computationally hard, since it requires $|X|$ forward passes to find the loss for a single data point.

Simplified objective

$$\min_{\mathbf{w}_{\mathcal{T}'}} \sum_{X \in D} \sum_{i=1}^{|X|} \left(- \mathbb{I} \left[\begin{array}{c} \mathbf{x}_{i+1} \in \mathcal{V} \\ \arg \max_{p_{\text{aug}} \in \mathcal{T}} \end{array} \right] \log p_{\text{rank}}(\text{REJ}|\mathbf{x}_{\leq i}, \mathcal{T}_k) - \mathbb{I}[\mathbf{x}_{i+1} \in \mathcal{T}] \log p_{\text{rank}}(\mathbf{x}_{i+1}|\mathbf{x}_{\leq i}, \mathcal{T}_k) \right)$$

Intuition: an approximation via hard negative mining (“hard” prefixes $\mathbf{x}_{\leq i}$ are those where ToolkenGPT incorrectly predicts tool use).

- **Tool selection** on MetaTool dataset: the task is to retrieve a single tool given a query, so there is no need to use the proposed REJ tool.
- **Multistep reasoning** on GSM8K dataset: the task is to generate a numerical answer to a question using the four basic arithmetic operations as tools.
- **Embodied plan generation** on VirtualHome dataset. the task is to generate a sequence of actions given a question. We follow the setup of the tool selection task but now generate a contiguous sequence of actions.

Experimental Results

LLM	Tool model	MetaTool	GSM8K	VirtualHome	
		Rec@1	Match	Strict	Relaxed
Vicuna-7B	4-shot	-	16.2	0.04	0.2
	ToolkenGPT	0.623	16.9	0.62	0.72
	Toolken+	0.643	18.8	0.48	0.74
Vicuna-13B	4-shot	-	17.8	0.16	0.30
	ToolkenGPT	0.646	18.4	0.34	0.54
	Toolken+	0.662	19.1	0.58	0.66
Llama2-7B-chat	4-shot	-	12.7	0.08	0.24
	ToolkenGPT	0.642	11.7	0.44	0.66
	Toolken+	0.692	12.8	0.56	0.66
Llama2-13B-chat	4-shot	-	10.3	0.18	0.26
	ToolkenGPT	0.704	8.8	0.18	0.20
	Toolken+	0.733	9.6	0.54	0.58

- We have proposed an improvement for the ToolkenGPT approach of learning special token embeddings that adds a reject option and a reranking mechanism for tool selection.
- Our approach significantly improves the results via in-context learning by optimizing only toolken embeddings.
- In future work, we hope to extend the Toolken+ approach to other external tools and/or agents to further expand the capabilities of modern LLMs.



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